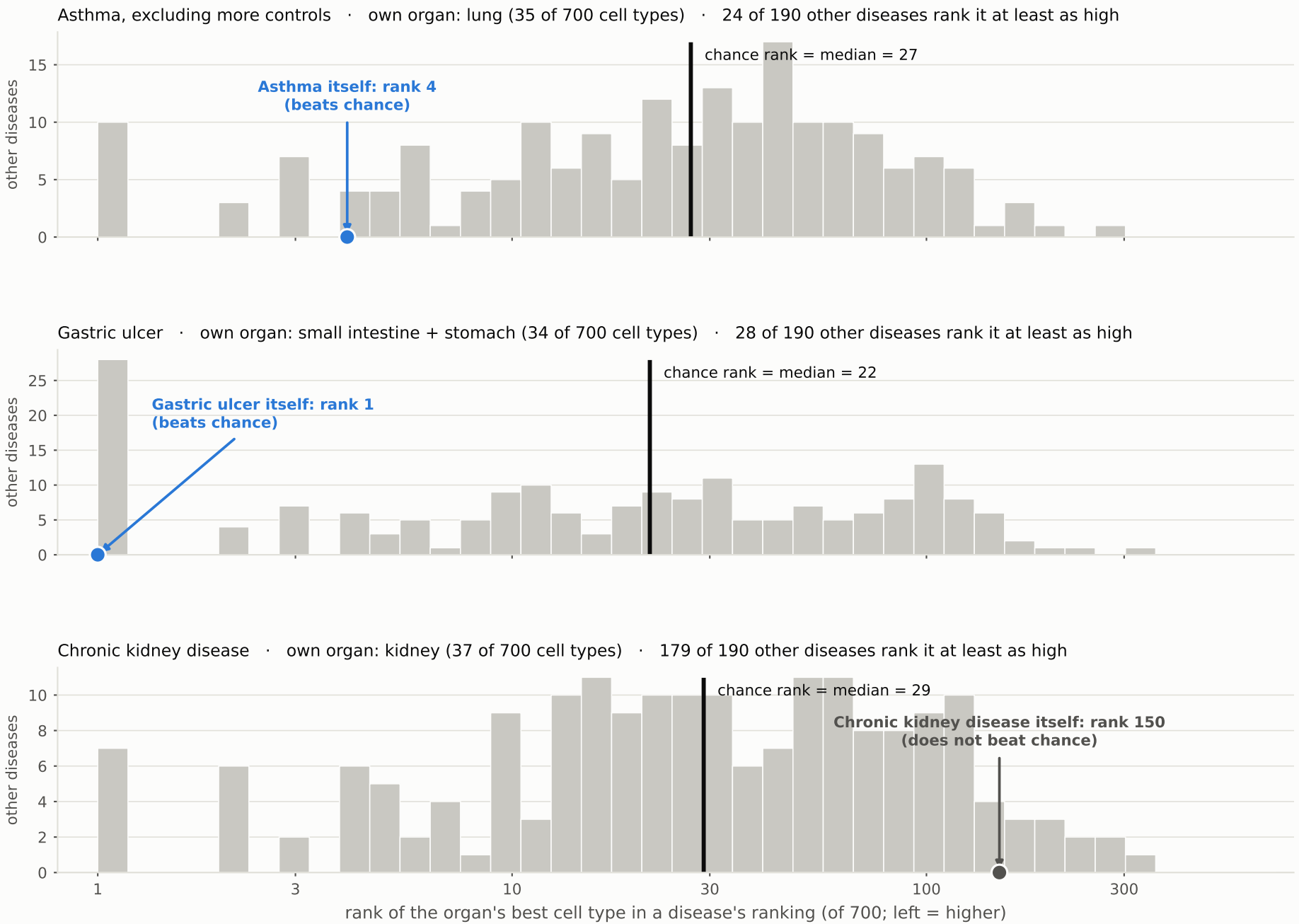
How the chance rank (the black tick in Figures 3 and 5) is computed

Gray: for each of the 190 OTHER diseases, where this disease's own-organ cell types first appear in that disease's ranking.
Black: the median of those 190 numbers = the chance rank. The disease beats chance if its own rank (dot) is left of the black line.
Because it is a median, a disease whose ranking says nothing about its organ would beat it about half the time: that is the '~50% expected'.